CSCI-338-Computer Science Theory
Fall 2025
HW 03

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Student Name:

This assignment is **due** on **October 10, 2025**. You will need to use Latex to generate a single pdf file and upload it under Assignment 03 on Canvas. There will be a penalty of not using Latex (to finish the assignment). This is not a group-assignment, so you must finish the assignment by yourself. **Figures can be hand drawn and inserted to the pdf.**

1 Problem 01

Prove if the following language is regular or not. $A = \{w | \ w$ contains the same number of 01 and 10 as substrings.$\}$. For example, $\epsilon, 1110101, 10001$ are in the language but 1010 is not.

Hint: If a language is regular, then there exists a DFA or NFA that recognizes the language. If language is not regular then you can use pumping lemma to prove that it is non-regular.

PICTURE INCLUDING DFA TO ANSWER QUESTION IS BELOW MY ANSWER TO PROBLEM 2. I COULDNT FIGURE OUT HOW TO GET IT IN THE RIGHT SPOT. Thus, the language is regular.

2 Problem 02

Prove that following language is not regular.

$$B = \{1^n | \ n \text{ is a prime.}\}$$

Proof: Assume B is regular. By the pumping lemma, there exists a pumping length p. Choose $s = 1^q$ where q is

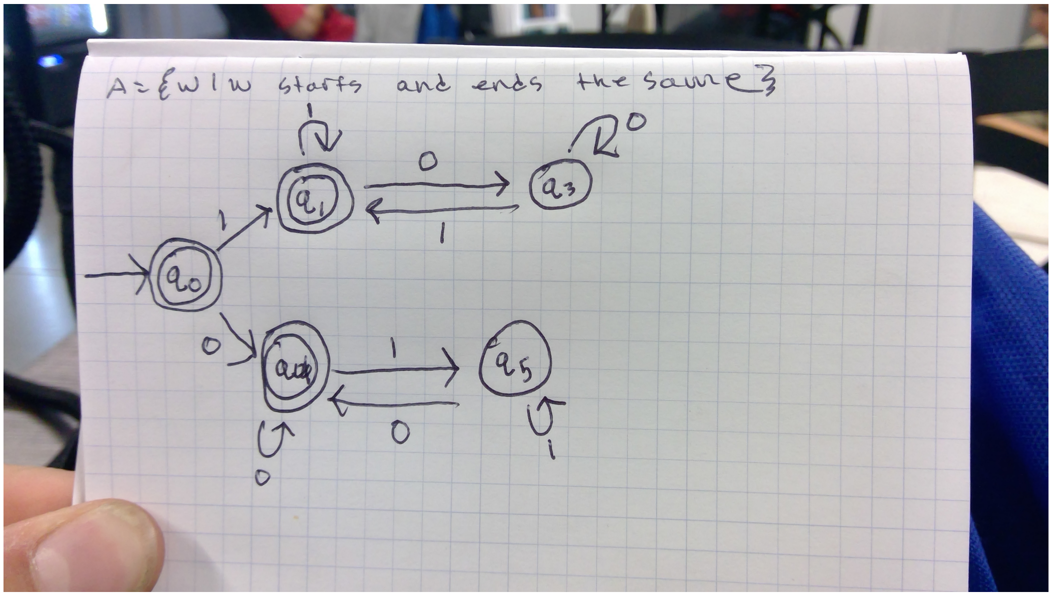

Figure 1: DFA.

a prime number greater than p. Split $s = xyz$ where $y = 1^k$ and $k > 0$. Pumping gives $xy^2z = 1^{q+k}$. Since $q + k$ may not be prime, the pumped string leaves the language, violating the lemma. Hence, B is not regular.

3 Problem 03

Suppose you want to prove that the language

$$C = \{www : w \in \Sigma^*\}.$$

is not regular using the pumping lemma.

You assume that the pumping length is p and choose the string $s = 0^p0^p0^p$ as a candidate that might violate the lemma.

Is this a suitable choice for proving that C is not regular. Explain your reasoning.

Proof: If we divide s into xyz with $|xy| \leq p$, then y lies entirely within the first block of 0's. Pumping y adds more 0's only to the first block, breaking the structure www. Therefore, the pumped string is not in C, violating the lemma, proving C is not regular.

4 Problem 04

Design context-free grammars for the following languages.

1. $A = \{a^n b^m \mid n \neq 2m\}$, where $\Sigma = \{a, b\}$.
2. $B = \{w \mid w \text{ contains at least three } 1s\}$, where $\Sigma = \{0, 1\}$.
3. $C = \{w \mid w = w^{\mathcal{R}}$, that is, w is a palindrome$\}$, where $\Sigma = \{0, 1\}$.
4. $D = \{a^n b^m \mid n = 3m\}$, where $\Sigma = \{a, b\}$.

1. $A = \{a^n b^m \mid n \neq 2m\}$

$$S \to aS \mid aaSbb \mid bS \mid ab \mid a.$$

2. $B = \{w \mid w \text{ contains at least three 1s}\}$

$$S \to A1A1A1A$$
$$A \to 0A \mid 1A \mid \epsilon$$

3. $C = \{w \mid w = w^R\}$ (palindromes)

$$S \to 0S0 \mid 1S1 \mid 0 \mid 1 \mid \epsilon$$

4. $D = \{a^n b^m \mid n = 3m\}$

$$S \to aaaSb \mid \epsilon$$

5 Problem 05

Decide whether the following grammar is ambiguous.

$S \to AB|aaB$

$A \to a|Aa$

$B \to b$

The grammar is ambiguous. Because the string "aab" can be derived in two ways: 1. $S \to AB \to AaB \to aab$ 2. $S \to aaB \to aab$. Since two distinct parse trees exist for the same string, the grammar is ambiguous.

6 Problem 06

Create the following CFG G to an equivalent PDA.

$R \to XRX \mid S$

$S \to aTb \mid bTa$

$T \to XTX \mid X \mid \epsilon$

$X \to a \mid b$

We build a PDA $P = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$ such that: , $Q = \{q\}$, $\Sigma = \{a, b\}$, $\Gamma = \{R, S, T, X\}$, $q_0 = q$, $Z_0 = R$

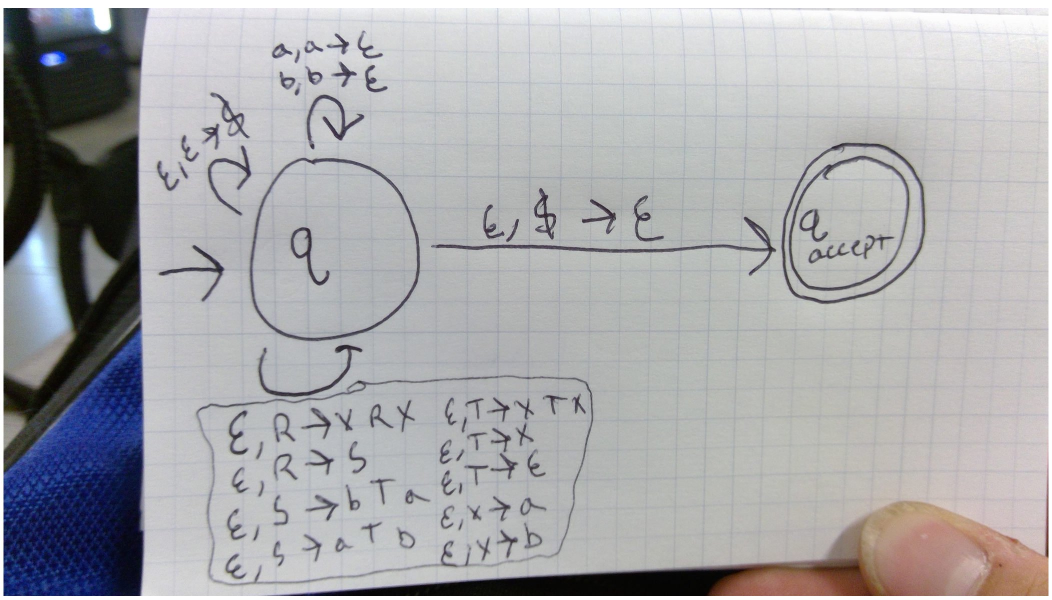

Figure 2: PDA.

7 Problem 06

Convert the following CFG into an equivalent CFG in Chomsky normal form, using the procedure given in Theorem 2.9 (in the book).

$A \to BAB \mid B \mid \epsilon$

$B \to 00 \mid \epsilon$

Given:

$$A \to BAB \mid B \mid \epsilon, \quad B \to 00 \mid \epsilon$$

Remove ϵ-rules: From $B \to \epsilon$ and $A \to \epsilon$,

$$A \to BAB \mid BA \mid AB \mid B \mid A \mid \epsilon$$

Eliminate unit rules: Replace $A \to B$ with $A \to 00 \mid \epsilon$.

Convert to CNF: Introduce variables:

$$C \to 00$$

Now CNF form is:

$$A \to BC \mid CB \mid AB \mid C$$

$$B \to C$$

$$C \to 00$$

8 Problem 07

Let $G = (V, \Sigma, R, S)$ be the following grammar. $V = \{S, T, U\}; \Sigma = \{0, \#\}$; and R is the set of rules:

$S \to TT \mid U$

$T \to 0T \mid T0 \mid \#$

$U \to 0U00 \mid \#$

Describe $L(G)$ in English.

- T generates all strings containing exactly one # anywhere in the middle of 0s (e.g., $00\#0, 0\#000$, etc.). - $S \to TT$ produces two such strings concatenated (so there are two # symbols total). - U generates strings with a single # surrounded by 0s, but the number of 0s on the right is double those on the left. Therefore:

$$L(G) = \{x\#y\#z \mid x, y, z \in 0^*\} \cup \{0^n\#0^{2n} \mid n \geq 0\}.$$